

The nine parity triples of the 3D DR Code

$$A_r \oplus B_r \oplus C_r = 0, \quad r = 0, \dots, 8$$


$$A_0 \oplus B_0 \oplus C_0 = 0$$


$$A_1 \oplus B_1 \oplus C_1 = 0$$


$$A_2 \oplus B_2 \oplus C_2 = 0$$


$$A_3 \oplus B_3 \oplus C_3 = 0$$


$$A_4 \oplus B_4 \oplus C_4 = 0$$


$$A_5 \oplus B_5 \oplus C_5 = 0$$


$$A_6 \oplus B_6 \oplus C_6 = 0$$


$$A_7 \oplus B_7 \oplus C_7 = 0$$


$$A_8 \oplus B_8 \oplus C_8 = 0$$

each block lies in exactly one triple; any one member is the XOR of the other two,
so an erased set decodes precisely when it meets every triple at most once